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PAPUA NEW GUINEA

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STAFF REPORT FOR THE 2018 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)[1]

Risk of external debt distress	*Moderate*	
Overall risk of debt distress	*Moderate*	
Granularity in the risk rating	*Limited space to absorb shocks*	
Application of judgment	Yes:	A single short-lived breach of debt service-to-revenue ratio is discounted.

The Debt Sustainability Analysis (DSA) indicates that Papua New Guinea (PNG) faces a moderate risk of debt distress based on an assessment of public external debt, with limited space to absorb shocks; and moderate overall risk of public debt distress. A single short-lived breach of debt service-to-revenue ratio is discounted because PNG has large revenue buffers from resource projects. This assessment, an improvement of overall public debt distress from that of December 2017, is based on PNG's upgraded debt-carrying capacity, and it continues to underscore the importance of strengthening the fiscal position – including through more cost-effective and longer-term financing – to improve debt dynamics.

[1] This DSA was prepared jointly with the World Bank, in accordance with the revised Debt Sustainability Framework for low-income countries approved by the Executive Boards of the IMF and the IDA in December 2017. The PNG's Composite Indicator score based on the October World Economic Outlook and 2017 CPIA measure indicates that the debt-carrying capacity is 'medium'. Thresholds for external debt-burden indicators are set at levels compatible with the new framework: PV PPG debt/exports (180), PV PPG debt/GDP (40), debt service/exports (15) and debt service to revenue (18). Note that the PNG is classified as an IDA blend economy.

PUBLIC DEBT COVERAGE

1. The Papua New Guinea (PNG)'s coverage of public sector debt is the central government and local government debt. The debt figure does not capture government guaranteed debts of state-owned enterprises (SOEs). The authorities are reviewing government guarantees on SOEs, which are currently estimated at 1.4 billion Kina, equivalent to about 2 percent of GDP, although the magnitude of the overall stock of SOE debt is uncertain due to lack of data. Thus, the contingent liability shock equivalent to 2 percent of GDP is included in this analysis to account for government guaranteed debt.

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

	The country's coverage of public debt	The central, state, and local governments			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
4	PPP	35	percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

2. PNG's public and external debt burdens remain moderate relative to peer countries, but debt and potential risks are increasing. Unanticipated revenue declines due to commodity price falls and difficulties in controlling government expenditure, especially the teachers' payrolls, in recent years have led to larger budget deficits and higher public debt. While the ratio of total public debt to GDP stands at relatively moderate levels, at 37.5 percent as of end-2017, government debt has almost doubled from 19 percent of GDP in 2012, and interest payments have tripled. Furthermore, liquidity risks are increasing as Treasury bills (less than 1 year to maturity) increased as a share of total domestic financing, from 45 percent in 2012, to 54 percent in 2017. In addition, the budget remains exposed to unfunded superannuation liabilities, relating to pensions, estimated at around 2 billion Kina (3 percent of 2017 GDP). The budget deficit financing turned to external sources as commercial banks and superannuation funds are nearing internal limits for holding government securities. The recent increase in external debt is mainly due to accelerated borrowing from non-Paris Club bilateral and commercial creditors (Text Table 1). The share of Chinese loans (the sum of loans by the Bank of China and the Export-Import Bank of China) has increased sharply from 1.2 percent of total public external debt in 2010 to 23.5 percent in 2017. The syndicated loans from Credit Suisse have also increased the public external debt since 2016. Two large creditors for external public debt at end-2017 were the Asian Development Bank and the Export-Import Bank of China (Chart 1). In addition, liquidity risks remain high as the maturity of government domestic financing has shortened. Private external debt has declined continuously since the liquified natural

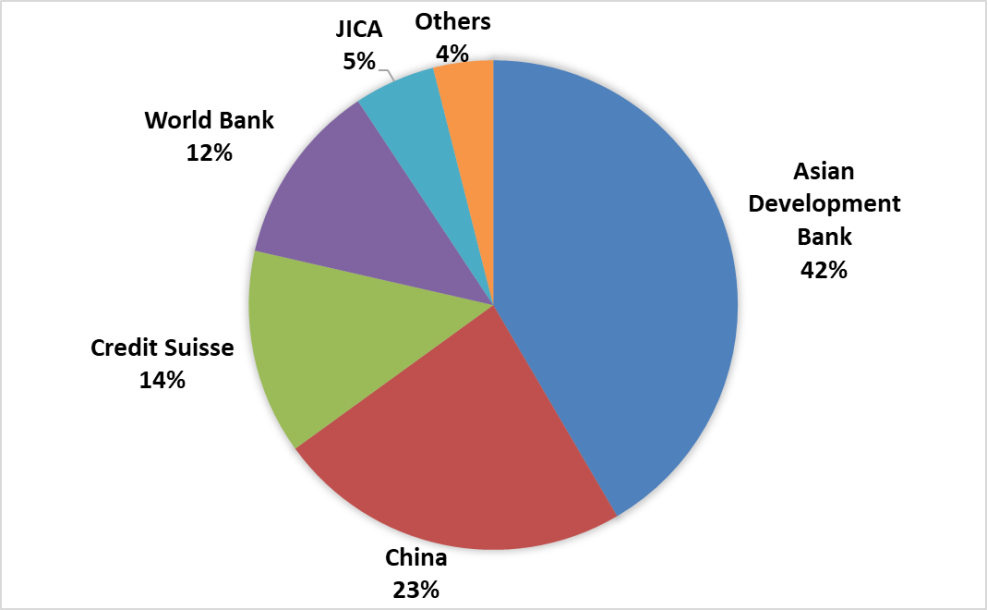
gas (LNG) sector started to repay its debt, while public and publicly-guaranteed (PPG) external debt risk remains moderate. Since a large current account surplus, which reflects LNG exports, does not bring enough foreign exchange (FX) to the country due to the project agreement, increased PPG external debt poses a risk to debt service. Namely, a significant residual on external financial flows is the money transfer via offshore accounts by resource companies.

Text Table 1: External Public Debt by Creditor Groups
(In millions of U.S. dollars)

	2015	2016	2017
Multilateral	1,086	1,164	1,272
Bilateral Paris Club	113	126	131
Bilateral Non-Paris Club	299	408	538
Commercial	0	229	348
Total	1,498	1,928	2,289

Source: Papua New Guinea Department of Treasury

Chart 1: External Public Debt by Creditors in 2017



BACKGROUND ON MACRO FORECASTS

3. Over the medium term, PNG's growth prospects and current account developments will be heavily influenced by its extractive sector and the commodity price outlook. Box 1 summarizes the medium-term macroeconomic framework underlying this Debt Sustainability Analysis (DSA) update, and Text Table 2 compares previous and current DSAs. The debt to GDP ratio differs from the authorities' numbers in the Mid-Year Economic and Fiscal Outlook Report and the Medium Term Fiscal Strategy (MTFS) because (1) new lower official 2015 GDP estimates published by the National Statistical Office and (2) effects of change in exchange rate on public external debt are incorporated in this DSA. Real GDP growth is expected to be zero in 2018, following an earthquake early in the year, but long-term growth has been revised up by 0.8 percentage points annually, reflecting a view for higher commodity prices than that which characterized the 2017 DSA. The current account surplus is boosted by steady commodity export growth. Nonetheless, the current account surplus is projected to decline as imports recover. The primary fiscal balance is projected to improve from a deficit of 0.5 percent of GDP in 2018 to a surplus of 0.8 percent by the end of the forecast period assuming the implementation of the Medium Term Revenue Strategy. Tax collection so far this year is up substantially from last year and well above budget projections. This reflects strong tax compliance efforts as well as the impact of higher commodity prices on resource revenues. We assume that this revenue mobilization efforts continue over time. Also, a new expenditure ceiling on personnel emoluments costs in the MTFS is assumed to avoid expenditure slippages throughout the projection period. The realism tool in Figure 4 shows that this assumption on fiscal adjustment is realistic. The baseline includes the issuance of a US$500 million 10-year sovereign bond with a yield of 8.375 percent. The funds are intended to be used to finance the deficit, reduce the government's short-term domestic debt, and reduce the large backlog of unfilled FX orders by the private sector. The possibility of significant capacity expansion in the resource sector poses an important upside risk to the baseline.

Text Table 2: Macroeconomic Forecast and Assumptions

	Real GDP Growth (percent)		GDP Deflator Inflation (percent)		Current Account (percent of GDP)		Primary Balance (percent of GDP)	
Year	Previous	Current	Previous	Current	Previous	Current	Previous	Current
2015	8.0	5.3	4.0	-4.4	13.3	11.8	-2.2	-3.0
2016	2.4	1.6	3.4	2.7	15.2	23.7	-2.6	-3.1
2017	2.2	2.5	2.0	7.1	13.9	23.9	-0.7	-0.4
2018	2.5	0.0	1.8	8.1	14.4	22.8	-0.5	-0.5
2019	2.3	3.8	2.3	2.4	13.8	23.0	-0.7	0.2
2020	2.9	3.1	2.4	2.0	12.8	21.4	-0.1	0.8
2021	3.2	3.4	2.4	2.4	12.3	20.3	-0.1	1.3
2022	3.1	3.5	2.4	2.5	11.8	19.2	-0.3	1.4
2023	3.1	3.8	2.4	1.2	11.3	18.4	-0.2	1.4

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

4. PNG's composite indicator index as calculated under the revised LIC-DSF has improved with PNG's debt-carrying capacity now classified as medium. Prior to this update, initial settings for debt-carrying capacity in the revised framework were set at that of the previous DSA vintage, where PNG was classified as 'weak'. The current composite indicator is calculated based on the October 2018 World Economic Outlook and 2017 CPIA index. The medium classification implies that the threshold for the present value of total public debt has increased from 35 percent to 55 percent of GDP. Since PNG is a resource-rich economy relying on commodity exports, commodity price shocks are included in the tailored stress tests.

EXTERNAL DSA

5. The PPG external debt ratios stay below the indicative thresholds under the baseline scenario, whilst stress tests support moderate risk of external debt distress. The baseline scenario is affected by planned increases in public external debt from 11.3 percent of GDP in 2017 to 15.3 percent in 2018. Although the historical shock scenario (Figure 1) shows substantial threshold breaches, as has been the case in past DSAs, this scenario is not considered indicative of future risks. This is because the historical scenario is grounded in large current account deficits recorded in the past associated with the construction phase for PNG LNG project. The PNG LNG-related debt was largely backed by expected cash flows from the LNG project. In contrast, the PPG external debt service-to-revenue ratio shows a breach in the baseline scenario, reflecting the servicing of PPG external debt. However, this is a 1-year and minimal breach and is discounted from the analysis (according to DSA guidelines). This assessment is also supported by PNG's large revenue buffers from future resource projects. In addition, the most extreme shock scenario, which corresponds to a case with shocks to export growth, shows a breach for the ratio of the present value of PPG external debt-to-GDP, while a shock to the PNG exchange rate also illicit a beach of threshold for debt-service to revenue. These outturns support the moderate-risk assessment. The relatively weak non-resource sector makes the economy more vulnerable to external shocks. But there is an upside risk that imports recover faster than the baseline scenario if a new resource project starts and triggers a boost to the economy, as the historical shock scenario of PNG LNG construction phase indicates.

OVERALL RISK OF PUBLIC DEBT DISTRESS

6. PNG's total public debt-to-GDP ratio is projected to remain below the benchmark, while it breaches the threshold under the stress test, indicating moderate risk. In the baseline, the PV of public debt-to-GDP ratio decreases towards 34 percent of GDP over the medium term,

below the benchmark (Figure 2). However, since PNG is a commodity exporter, in the alternative scenarios with shocks to exports or shocks to commodity price, the public debt-to-GDP ratio will breach the benchmark in the medium term (Table 4). In addition, the most extreme shock for the debt service-to-revenue ratio is the commodity price shock (Figure 2 and Table 4). The negative commodity price shock could reduce commodity export revenues, resulting in lower mining taxes and dividends to the government. Furthermore, liquidity risks are increasing as elaborated in the background of debt section, and unfunded superannuation liabilities represent an implicit fiscal risk for PNG.

RISK RATING AND VULNERABILITIES

7. PNG's risk of external public debt distress remains moderate and debt distress of overall public debt is also moderate. The moderate risk of overall public debt distress reflects in part the upgrade of debt-carrying capacity from weak to medium, supported by recent macroeconomic and policy performance. The moderate PPG external debt distress is based on the assumption of primary fiscal surpluses generated from fiscal consolidation efforts by the government. Granular analysis in Figure 5 shows that PNG has limited space to absorb shocks as the debt service-to-revenue ratio almost reaches the respective threshold even if the short-lived breach in 2020 is discounted. Against this background, implementation of a sound debt management strategy, to better capture existing government liabilities and guarantees grows in importance.

AUTHORITIES' VIEWS

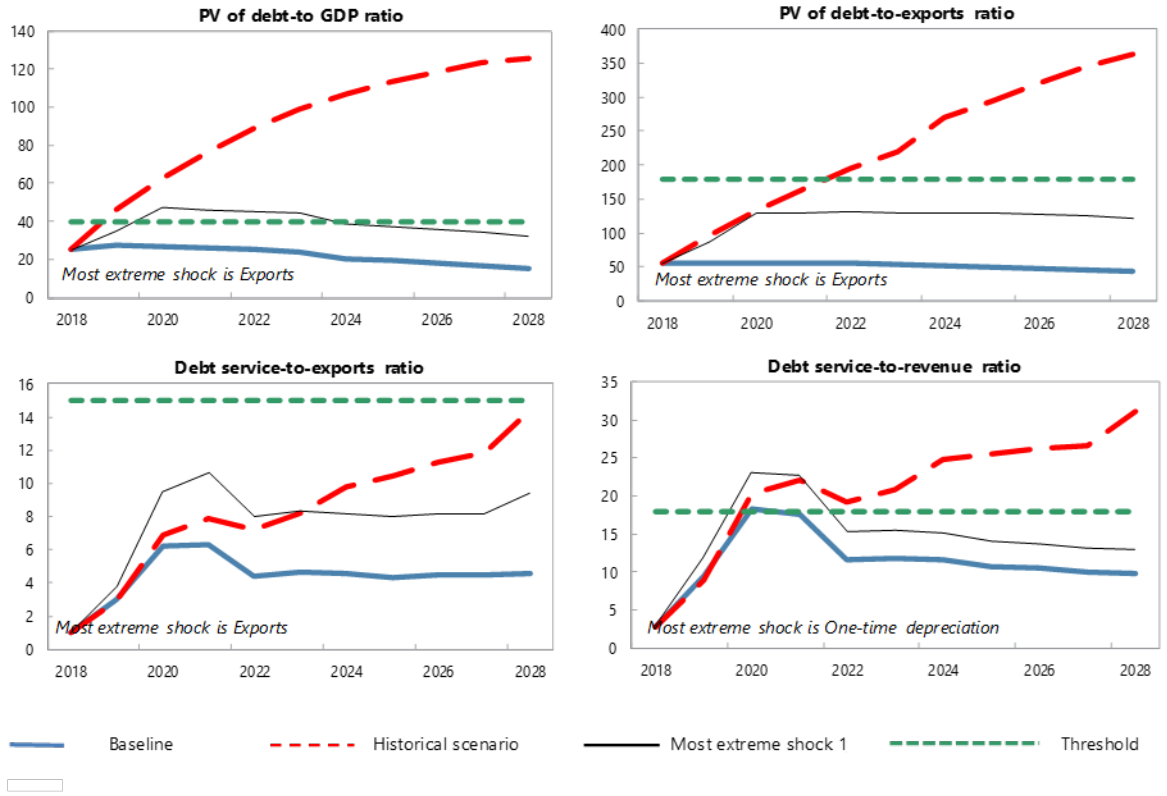
8. The authorities took note of an increasing risks to debt sustainability. They also recognized three statistical issues. First, it is important to agree on the new official GDP numbers among the National Statistical Office, the government, and the central bank to precisely measure the debt-to-GDP ratio. Second, they recognized that it is also important to include the change in exchange rates when calculating the external debt stock in domestic currency. Third, the calculating exercise for government guarantees on SOEs is critical to know the contingent liabilities. Current bond issue and other external financing will give the government flexibility to address domestic FX imbalances and problems in securities markets by retiring short-term government securities.

Box 1. Macroeconomic Assumptions Underlying the DSA Update

The macroeconomic assumptions underpinning this DSA are largely unchanged contrasted with the 2017 DSA, except for inclusion of a sovereign bond issuance and recent high commodity prices.

Macro Indicators	Last Historic Year	First Forecast Year	Medium- and Long-Term Averages
GDP growth	Newly published lower 2015 nominal GDP series are reflected. The downward revision from previous estimates arose from sectoral real growths and their deflators, which incorporated an improved methodology to take better account of differences between input and output prices. The single extrapolation method introduced improved on the previous single deflation method (where input and output prices were assumed to move identically).	The impact of earthquakes occurred in early 2018 is incorporated in to the framework. This lowered the real growth rate from 2.5 percent in the 2017 DSA to 0 percent in this DSA.	Real GDP growth is projected to average 3.9 percent in the medium/long run, an increase from the 3.1 percent long term average growth rate used for the 2017 DSA. This upward revision was mainly based on better sentiment for the mining, petroleum and gas sectors coupled with an improvement in commodity prices (e.g. gold, copper, silver, and nickel).
Inflation (GDP deflator)	The actual inflation rate (7.1 percent) in 2017 was substantially higher than the projection in the 2017 DSA (1.4 percent) due to higher commodity prices.	Inflation is projected at around 8 percent in 2018 reflecting high fuel and commodity prices (revised from 0.6 percent in the 2017 DSA).	Inflation is expected to converge to core inflation (3 percent) in the long term, assuming a relatively stable exchange rate.
Current Account	The estimated current account to GDP ratio in 2017 is revised upwards from 13.9 percent in the 2017 DSA to 23.9 percent, reflecting higher resource exports and lower GDP.	High fuel and commodity prices will lead to steady resource exports, albeit the earthquake impact.	Gradual erosion of the current account surplus is projected as FX backlog will be cleared mainly in 2019, and imports recover.
Fiscal Position	The primary deficit in 2017 was 0.4 percent of GDP, a bit smaller than 0.7 percent estimated in the 2017 DSA.	The primary fiscal deficit is estimated to be 0.5 percent of GDP in 2018. Sovereign bond issuance is included in 2018, and its redemption will take place in 2028. The interest rate is assumed to be 8.375 percent.	The primary deficit will gradually improve to 0.8 percent of surplus in the long run, assuming fiscal consolidation by improving tax compliance and controlling government expenditures better.

Figure 1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2018-2028



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	30	30
Avg. grace period	8	8

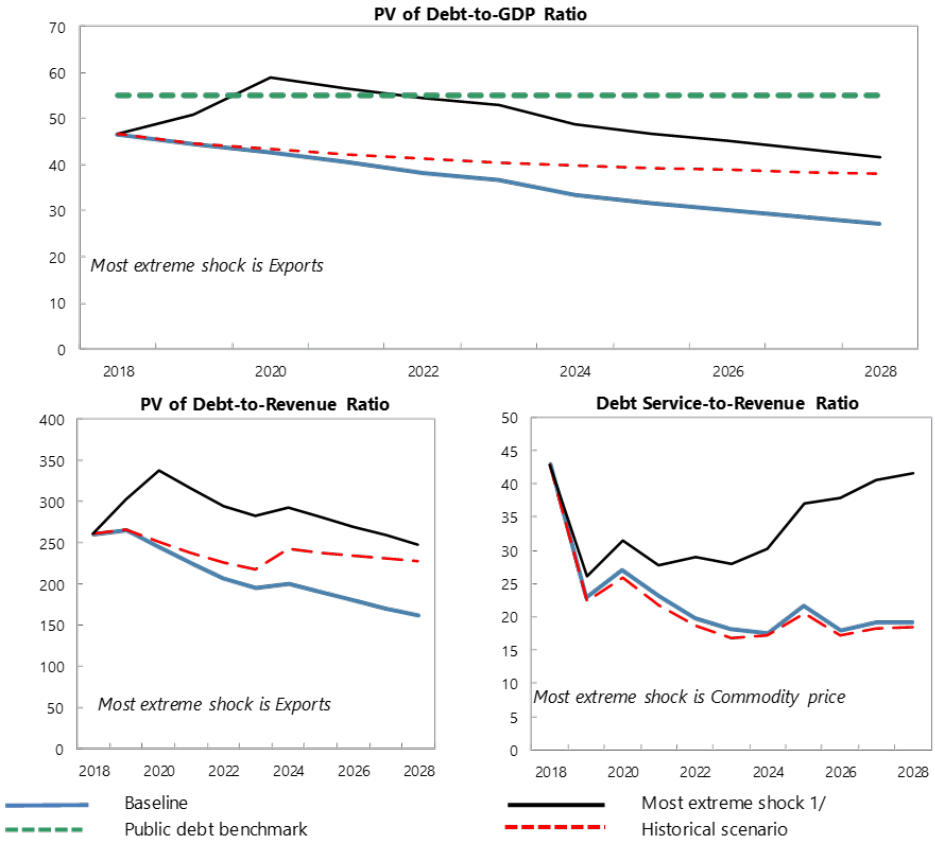
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2018-2028



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	47%	47%
Domestic medium and long-term	37%	37%
Domestic short-term	16%	16%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
Avg. maturity (incl. grace period)	30	30
Avg. grace period	8	8
Domestic MLT debt		
Avg. real interest rate on new borrowing	7.2%	7.2%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	4%	4.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

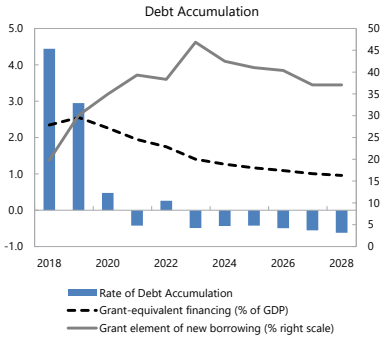
1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

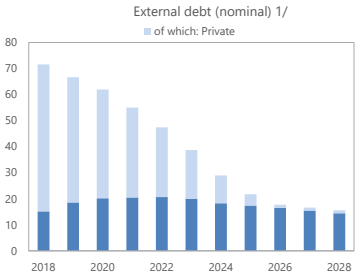
Table 1. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2015-2038

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**97.8**	**96.6**	**82.1**	**71.6**	**66.7**	**61.9**	**55.0**	**47.4**	**38.7**	**15.6**	**6.0**	**70.7**	**40.2**
of which: public and publicly guaranteed (PPG)	**7.9**	**10.2**	**11.3**	**15.2**	**18.6**	**20.2**	**20.5**	**20.6**	**20.0**	**14.5**	**5.3**	**7.8**	**17.9**
Change in external debt	1.5	-1.3	-14.4	-10.5	-4.9	-4.7	-7.0	-7.6	-8.7	-1.1	-0.8		
Identified net debt-creating flows	**-1.8**	**-15.1**	**-28.4**	**-26.4**	**-26.6**	**-24.7**	**-23.5**	**-22.2**	**-21.2**	**-11.1**	**-5.6**	**2.0**	**-19.8**
Non-interest current account deficit	**-14.5**	**-27.2**	**-27.2**	**-25.7**	**-26.4**	**-25.4**	**-24.0**	**-22.0**	**-20.8**	**-12.2**	**-5.3**	**3.5**	**-19.7**
Deficit in balance of goods and services	-15.8	-26.6	-26.7	-25.9	-27.1	-26.0	-24.9	-23.8	-22.9	-15.7	-8.3	**0.9**	**-21.7**
Exports	38.5	46.3	49.0	46.0	48.4	47.7	46.7	45.6	44.9	34.5	24.5		
Imports	22.7	19.7	22.3	20.0	21.3	21.7	21.8	21.8	22.0	18.8	16.3		
Net current transfers (negative = inflow)	-0.8	-1.5	-0.8	-1.2	-0.6	-0.5	-0.6	-0.7	-0.8	-0.1	0.2	**-1.0**	**-0.5**
of which: official	-1.5	-1.7	-1.0	-1.0	-0.9	-0.8	-0.9	-0.9	-0.9	-0.8	-0.7		
Other current account flows (negative = net inflow)	2.1	1.0	0.2	1.4	1.3	1.1	1.5	2.5	2.9	3.6	2.7	**3.6**	**2.6**
Net FDI (negative = inflow)	**-1.3**	**0.4**	**1.6**	**-3.7**	**-1.0**	**-1.2**	**-1.1**	**-1.1**	**-1.1**	**0.7**	**-0.5**	**-0.8**	**-1.1**
Endogenous debt dynamics 2/	**14.1**	**11.7**	**-2.8**	**3.0**	**0.8**	**2.0**	**1.7**	**0.9**	**0.7**	**0.4**	**0.1**		
Contribution from nominal interest rate	2.8	3.4	3.4	3.0	3.4	4.0	3.7	2.7	2.5	1.0	0.4		
Contribution from real GDP growth	-5.7	-1.7	-2.3	0.0	-2.6	-2.0	-2.0	-1.8	-1.7	-0.6	-0.3		
Contribution from price and exchange rate changes	17.0	10.0	-3.9	...	...	...	...	...	...	...	...		
Residual 3/	**3.3**	**13.8**	**14.0**	**15.9**	**21.7**	**19.9**	**16.5**	**14.6**	**12.5**	**10.1**	**4.8**	**4.2**	**13.8**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**22.0**	**25.1**	**27.1**	**26.9**	**25.7**	**25.1**	**23.9**	**15.0**	**4.4**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**44.9**	**54.7**	**56.0**	**56.3**	**55.0**	**55.0**	**53.3**	**43.5**	**17.7**		
PPG debt service-to-exports ratio	**1.0**	**1.3**	**1.4**	**1.0**	**3.0**	**6.2**	**6.3**	**4.4**	**4.6**	**4.6**	**3.4**		
PPG debt service-to-revenue ratio	**2.3**	**4.0**	**4.3**	**2.8**	**9.4**	**18.2**	**17.6**	**11.6**	**11.8**	**9.9**	**4.9**		
Gross external financing need (Million of U.S. dollars)	-887.6	-1100.2	-2349.2	-2874.8	-3557.3	-3783.8	-3435.7	-3201.0	-2860.0	-3443.8	-2782.7		
Key macroeconomic assumptions													
Real GDP growth (in percent)	5.3	1.6	2.5	0.0	3.8	3.1	3.4	3.5	3.8	3.8	3.9	**5.1**	**3.3**
GDP deflator in US dollar terms (change in percent)	-15.0	-9.2	4.2	5.2	-0.2	-0.5	-0.2	-0.1	-1.0	1.2	1.2	**3.5**	**1.6**
Effective interest rate (percent) 4/	2.6	3.2	3.8	3.8	4.9	6.2	6.2	5.2	5.3	6.3	5.5	**3.4**	**5.4**
Growth of exports of G&S (US dollar terms, in percent)	-11.4	10.8	13.1	-1.3	9.0	1.2	0.9	1.0	1.2	1.7	0.1	**8.7**	**1.7**
Growth of imports of G&S (US dollar terms, in percent)	-40.1	-20.1	21.2	-5.5	10.1	4.6	3.4	3.7	3.5	3.5	3.7	**4.5**	**3.4**
Grant element of new public sector borrowing (in percent)	...	...	...	19.8	30.1	34.9	39.3	38.3	46.9	37.1	#DIV/0!	**...**	**37.0**
Government revenues (excluding grants, in percent of GDP)	17.8	15.2	15.4	16.4	15.6	16.2	16.7	17.3	17.6	16.0	17.1	**10.6**	**16.2**
Aid flows (in Million of US dollars) 5/	490.4	946.6	879.0	492.7	404.2	486.2	509.9	464.0	392.2	337.4	298.2		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.3	2.6	2.3	1.9	1.7	1.4	1.0	...	**...**	**1.6**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	39.2	45.8	52.9	62.7	64.6	81.3	83.4	...	**...**	**68.6**
Nominal GDP (Million of US dollars)	20,639	19,028	20,334	21,390	22,165	22,731	23,442	24,233	24,884	34,597	56,889		
Nominal dollar GDP growth	-10.5	-7.8	6.9	5.2	3.6	2.6	3.1	3.4	2.7	5.0	5.2	**8.6**	**5.0**
Memorandum items:													
PV of external debt 7/	...	...	92.8	81.6	75.2	68.6	60.1	51.9	42.6	16.1	5.0		
In percent of exports	...	...	189.5	177.5	155.5	143.8	128.9	113.8	94.9	46.8	20.6		
Total external debt service-to-exports ratio	30.0	45.4	28.7	34.7	23.5	21.0	22.5	21.6	23.2	4.6	3.4		
PV of PPG external debt (in Million of US dollars)			4472.8	5376.1	6006.9	6112.2	6015.5	6076.5	5957.6	5199.7	2477.1		
(PVt-PVt-1)/GDPt-1 (in percent)				4.4	2.9	0.5	-0.4	0.3	-0.5	-0.6	-0.6		
Non-interest current account deficit that stabilizes debt ratio	-16.1	-25.9	-12.8	-15.2	-21.5	-20.7	-17.1	-14.4	-12.1	-11.1	-4.4		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

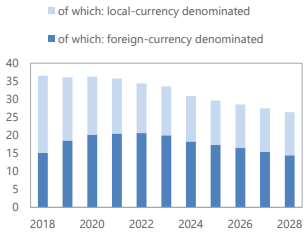
Table 2. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-2038

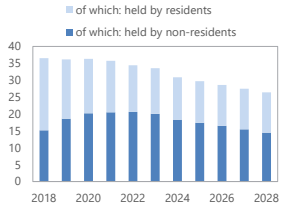
(In percent of GDP unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**32.2**	**37.7**	**37.5**	**36.5**	**36.1**	**36.3**	**35.7**	**34.4**	**33.5**	**26.4**	**17.5**	**20.9**	**32.3**
of which: external debt	**7.9**	**10.2**	**11.3**	**15.2**	**18.6**	**20.2**	**20.5**	**20.6**	**20.0**	**14.5**	**5.3**	**7.8**	**17.9**
Change in public sector debt	5.2	5.5	-0.2	-1.0	-0.4	0.2	-0.6	-1.3	-0.9	-1.1	-0.7		
Identified debt-creating flows	**5.7**	**4.4**	**-0.4**	**-1.0**	**-0.4**	**0.2**	**-0.6**	**-1.3**	**-0.9**	**-1.0**	**-0.7**	**1.4**	**-1.0**
Primary deficit	**3.0**	**3.1**	**0.4**	**0.5**	**-0.2**	**-0.8**	**-1.3**	**-1.4**	**-1.4**	**-1.0**	**-0.8**	**1.7**	**-0.9**
Revenue and grants	19.3	17.6	17.6	17.9	16.8	17.4	17.9	18.4	18.7	16.8	17.6	**11.7**	**17.3**
of which: grants	1.4	2.4	2.2	1.4	1.3	1.2	1.2	1.2	1.1	0.8	0.5		
Primary (noninterest) expenditure	22.3	20.7	18.0	18.4	16.7	16.6	16.6	17.0	17.4	15.8	16.8	**13.4**	**16.4**
Automatic debt dynamics	**2.6**	**1.3**	**-0.8**	**-1.5**	**-0.2**	**0.9**	**0.7**	**0.1**	**0.5**	**0.0**	**0.1**		
Contribution from interest rate/growth differential	1.3	0.9	-0.6	-1.1	-0.4	0.6	0.4	-0.2	-0.1	-0.1	0.1		
of which: contribution from average real interest rate	2.7	1.4	0.4	-1.1	0.9	1.7	1.6	1.0	1.2	0.9	0.8		
of which: contribution from real GDP growth	-1.4	-0.5	-0.9	0.0	-1.3	-1.1	-1.2	-1.2	-1.3	-1.0	-0.7		
Contribution from real exchange rate depreciation	1.3	0.3	-0.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-0.5**	**1.1**	**0.1**	**-0.4**	**0.2**	**0.3**	**0.3**	**0.3**	**0.6**	**0.1**	**0.0**	**1.2**	**0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**48.5**	**46.7**	**44.6**	**42.7**	**40.5**	**38.2**	**36.7**	**27.1**	**16.6**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**275.5**	**261.0**	**265.0**	**245.5**	**225.5**	**207.2**	**195.9**	**161.3**	**94.3**		
Debt service-to-revenue and grants ratio 3/	**91.8**	**125.4**	**117.9**	**42.9**	**22.9**	**27.1**	**23.2**	**19.9**	**18.2**	**19.2**	**21.3**		
Gross financing need 4/	20.7	25.2	21.2	8.2	3.7	4.0	2.9	2.3	2.0	2.2	2.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	5.3	1.6	2.5	0.0	3.8	3.1	3.4	3.5	3.8	3.8	3.9	**5.1**	**3.3**
Average nominal interest rate on external debt (in percent)	1.5	2.7	2.7	3.5	8.5	11.7	10.6	7.2	7.3	6.9	6.2	**1.9**	**7.6**
Average real interest rate on domestic debt (in percent)	13.3	5.6	1.1	-4.8	-0.2	-0.2	-0.4	-0.1	1.2	1.5	4.5	**3.6**	**-0.5**
Real exchange rate depreciation (in percent, + indicates depreciation)	22.6	4.1	-2.1	...	...	...	...	...	...	...	...	**0.3**	**...**
Inflation rate (GDP deflator, in percent)	-4.4	2.7	7.1	8.1	2.4	2.0	2.4	2.5	1.2	3.0	3.0	**3.7**	**3.5**
Growth of real primary spending (deflated by GDP deflator, in percent)	-8.4	-5.7	-10.7	2.0	-5.8	2.9	3.4	5.8	5.9	4.3	4.6	**3.1**	**2.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-2.1	-2.4	0.7	1.5	0.2	-0.9	-0.7	-0.1	-0.5	0.0	-0.1	**-1.3**	**0.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018-2028

(In percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	25	27	27	26	25	24	21	19	18	16	15
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	25	**46**	**63**	**77**	**89**	**99**	**107**	**113**	**118**	**123**	**126**
B. Bound Tests											
B1. Real GDP growth	25	29	31	30	29	28	24	22	21	19	17
B2. Primary balance	25	28	30	29	29	28	25	24	23	21	20
B3. Exports	25	35	**47**	**46**	**45**	**44**	39	37	36	34	32
B4. Other flows 3/	25	29	30	29	28	27	23	22	21	19	18
B5. One-time 30 percent nominal depreciation	25	34	37	35	35	33	29	27	25	23	22
B6. Combination of B1-B5	25	34	37	36	35	34	29	28	26	24	23
C. Tailored Tests											
C1. Combined contingent liabilities	25	29	30	28	28	27	24	23	22	20	19
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	25	33	39	39	40	40	36	35	34	33	31
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	55	56	56	55	55	53	52	50	48	46	44
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	55	95	132	164	**195**	**220**	**269**	**295**	**321**	**346**	**364**
B. Bound Tests											
B1. Real GDP growth	55	56	56	55	55	53	52	50	48	46	44
B2. Primary balance	55	59	63	62	63	62	62	62	61	60	58
B3. Exports	55	86	130	130	131	130	129	129	128	126	122
B4. Other flows 3/	55	59	63	62	62	60	59	58	56	54	51
B5. One-time 30 percent nominal depreciation	55	56	61	60	60	58	57	55	54	52	49
B6. Combination of B1-B5	55	72	69	78	78	77	75	74	72	70	67
C. Tailored Tests											
C1. Combined contingent liabilities	55	61	62	61	61	61	61	60	59	57	55
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	55	84	96	95	96	94	93	94	95	95	94
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	1	3	6	6	4	5	5	4	4	4	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	1	3	7	8	7	8	10	10	11	12	14
B. Bound Tests											
B1. Real GDP growth	1	3	6	6	4	5	5	4	4	4	5
B2. Primary balance	1	3	6	7	5	5	5	5	5	5	5
B3. Exports	1	4	10	11	8	8	8	8	8	8	9
B4. Other flows 3/	1	3	6	7	5	5	5	5	5	5	5
B5. One-time 30 percent nominal depreciation	1	3	6	6	5	5	5	5	5	5	5
B6. Combination of B1-B5	1	3	7	8	6	6	6	5	6	6	6
C. Tailored Tests											
C1. Combined contingent liabilities	1	3	6	6	5	5	5	5	5	5	5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	1	4	8	8	6	6	6	6	6	6	7
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	3	9	**18**	18	12	12	12	11	10	10	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	3	9	**20**	**22**	**19**	**21**	**25**	**25**	**26**	**27**	**31**
B. Bound Tests											
B1. Real GDP growth	3	10	**21**	**20**	14	14	13	12	12	12	11
B2. Primary balance	3	9	**18**	**18**	12	12	12	11	11	11	11
B3. Exports	3	10	**21**	**23**	16	16	16	15	14	14	15
B4. Other flows 3/	3	9	**18**	**18**	12	12	12	11	11	11	11
B5. One-time 30 percent nominal depreciation	3	12	**23**	**23**	15	15	15	14	14	13	13
B6. Combination of B1-B5	3	10	**21**	**21**	14	14	14	13	13	12	13
C. Tailored Tests											
C1. Combined contingent liabilities	3	9	**19**	**18**	12	12	12	11	11	11	11
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	3	11	**22**	**23**	16	16	15	14	14	13	15
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2018-2028

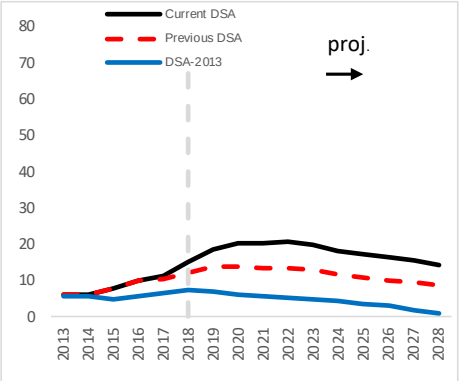
(In percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	47	45	43	40	38	37	33	32	30	29	27
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	47	45	43	42	41	40	39	39	38	38	38
B. Bound Tests											
B1. Real GDP growth	47	49	53	53	52	53	51	51	51	51	51
B2. Primary balance	47	48	50	48	46	44	40	39	37	35	34
B3. Exports	47	51	**59**	**57**	54	53	49	47	45	43	42
B4. Other flows 3/	47	46	46	44	41	40	36	34	33	31	30
B5. One-time 30 percent nominal depreciation	47	50	46	43	40	37	33	31	28	26	24
B6. Combination of B1-B5	47	47	46	41	39	37	34	32	30	29	27
C. Tailored Tests											
C1. Combined contingent liabilities	47	51	49	46	44	43	39	37	36	34	32
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	47	47	49	51	53	54	53	53	53	54	54
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	261	265	246	226	207	196	201	190	180	171	161
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	261	266	248	234	222	215	238	234	231	228	225
B. Bound Tests											
B1. Real GDP growth	261	290	301	290	281	280	303	303	303	303	304
B2. Primary balance	261	286	288	267	248	236	244	232	222	212	202
B3. Exports	261	303	338	316	295	282	293	281	270	260	248
B4. Other flows 3/	261	274	263	243	224	212	218	207	197	187	178
B5. One-time 30 percent nominal depreciation	261	296	268	241	217	201	201	186	171	157	144
B6. Combination of B1-B5	261	278	267	231	212	200	205	193	183	173	163
C. Tailored Tests											
C1. Combined contingent liabilities	261	300	279	259	240	227	234	223	213	203	193
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	261	315	314	315	307	304	327	319	319	320	321
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	43	23	27	23	20	18	18	22	18	19	19
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	43	22	26	22	19	17	17	21	17	18	19
B. Bound Tests											
B1. Real GDP growth	43	25	33	30	27	27	32	39	36	38	40
B2. Primary balance	43	23	32	31	24	25	30	32	23	24	25
B3. Exports	43	23	28	26	23	21	20	24	21	22	23
B4. Other flows 3/	43	23	27	24	20	19	18	22	18	20	20
B5. One-time 30 percent nominal depreciation	43	23	31	27	22	20	20	23	19	20	20
B6. Combination of B1-B5	43	22	27	25	20	19	19	23	19	19	20
C. Tailored Tests											
C1. Combined contingent liabilities	43	23	35	26	22	26	28	27	21	23	24
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	43	26	31	28	29	28	30	37	38	41	42
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

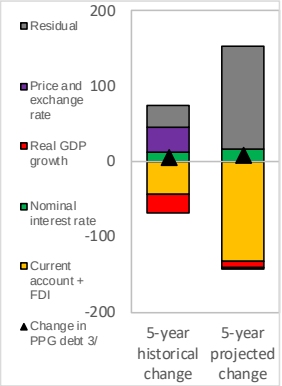
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Papua New Guinea: Drivers of Debt Dynamics – Baseline Scenario External Debt

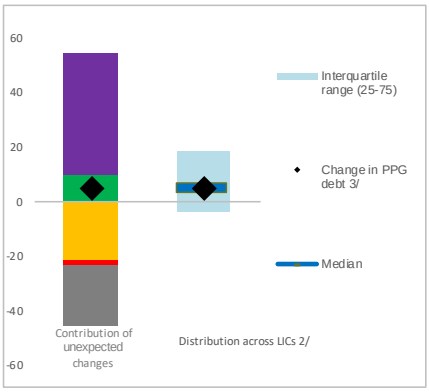
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

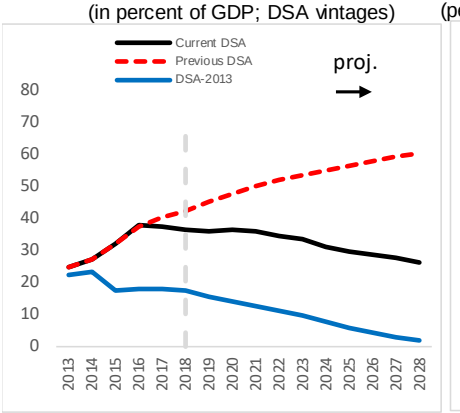


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

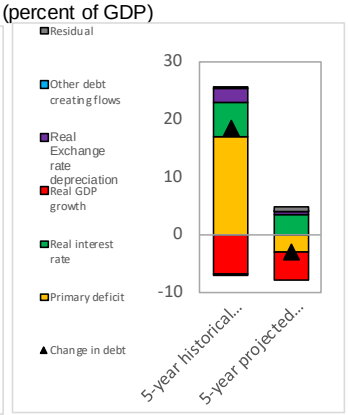


Public debt

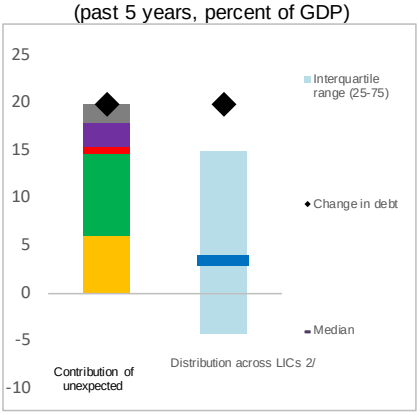
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



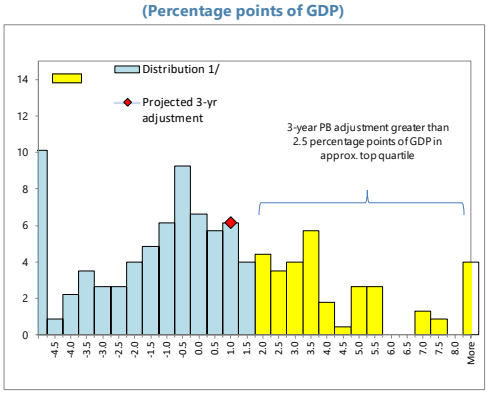
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

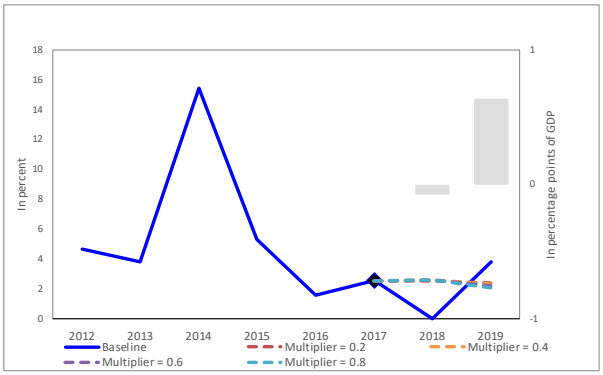
Figure 4. Papua New Guinea: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates

(% of GDP)

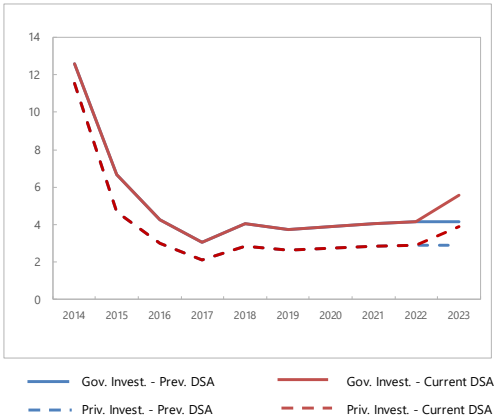
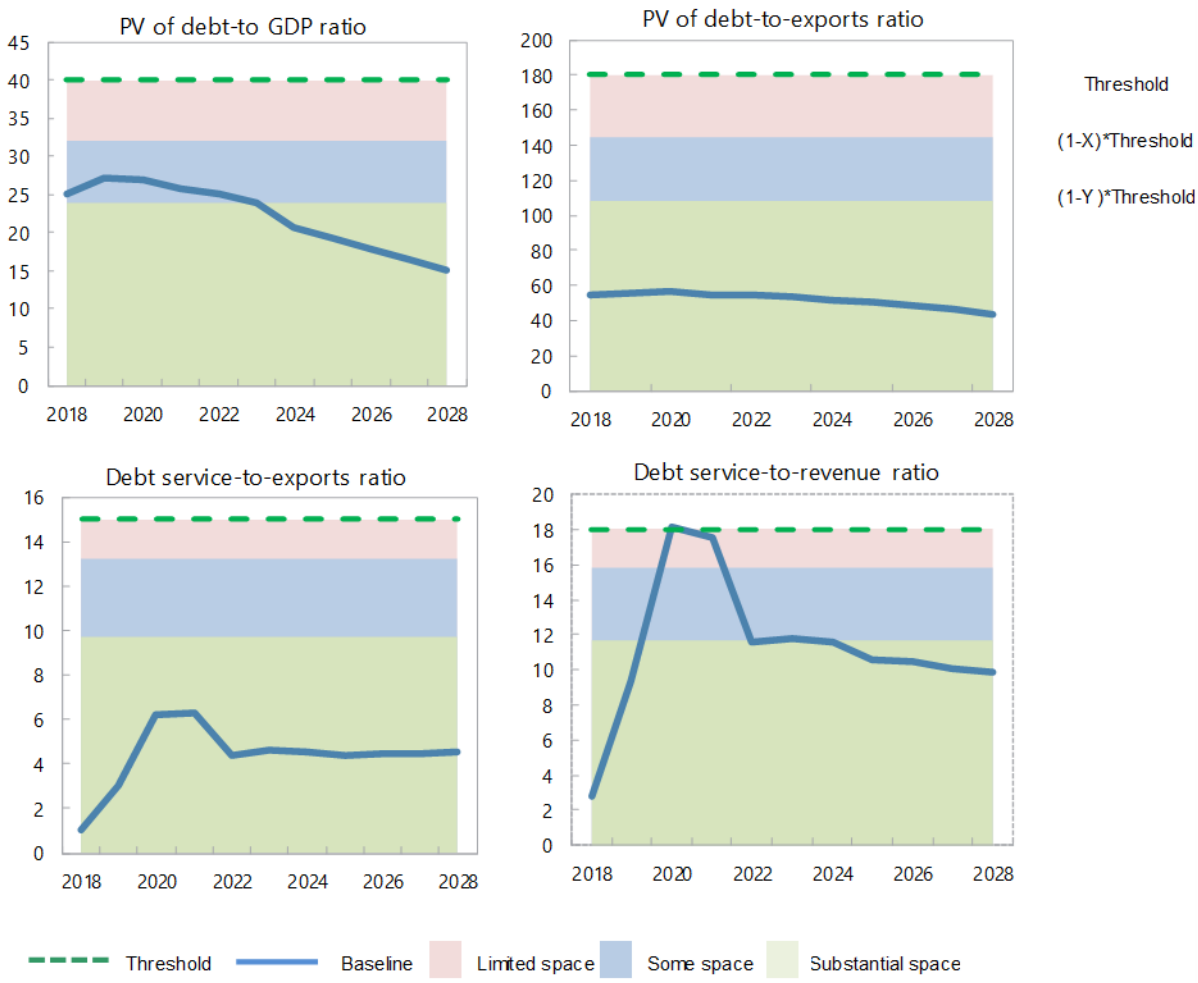


Figure 5. Papua New Guinea: Qualification of the Moderate Category, 2018-2028 1/



Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.